

XIAMEN
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COM102 – Programming Language C

Chapter 7: Input and Output (ii) – Lab

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Task 1

Write a C program that:

1. Prompts the user to enter **a sequence of integers** (one by one).
2. Continues reading integers **until the user enters a negative number**.
3. Opens a text file named **numbers.txt** in **write mode**.
4. Writes **only the non-negative integers** to the file, **one integer per line**, using file handling.
5. Stops writing when a negative number is encountered (the negative number must **not** be written to the file).
6. Closes the file.
7. Reopens the file in **read mode**, reads all the stored integers, and displays them on the screen.

Task 1 - Solution

```c
#include <stdio.h>
int main() {
    FILE *fp;
    int num;
    // Open file in write mode
    fp = fopen("numbers.txt", "w");
    if (fp == NULL) {
        printf("Error opening file!\n");
        return 1;
    }
    printf("Enter integers (negative number to stop):\n");
    // Read integers until a negative number is entered
    while (1) {
        scanf("%d", &num);

        if (num < 0) {
            break;  // stop input and writing
        }

        fprintf(fp, "%d\n", num);
    }
    fclose(fp);
    // Reopen file in read mode
    fp = fopen("numbers.txt", "r");
    if (fp == NULL) {
        printf("Error opening file for reading!\n");
        return 1;
    }
```

Task 2

Develop a C program that:

1. Prompts the user to enter their **full name** (first name + last name).
2. Extracts and stores **only the first name** (the characters before the first space).
3. Opens a text file named **firstname.txt** in **write mode** and writes the **first name character-by-character** to the file.
4. Closes the file.

Task 2 - Solution

```c
#include <stdio.h>
int main() {
    FILE *fp;
    char name[100];
    int i = 0;
    // Ask user to enter full name
    printf("Enter your full name: ");
    fgets(name, sizeof(name), stdin);
    // Open file in write mode
    fp = fopen("firstname.txt", "w");
    if (fp == NULL) {
        printf("Error opening file!\n");
        return 1;
    }
    // Write first name character by character (until space or newline)
    while (name[i] != '\0' && name[i] != '\n' && name[i] != ' ') {
        fputc(name[i], fp);
        i++;
    }
    fclose(fp);
    // Reopen file in read mode
    fp = fopen("firstname.txt", "r");
    if (fp == NULL) {
        printf("Error opening file for reading!\n");
        return 1;
    }
    // Read and display the first name
    printf("Saved first name: ");
    int ch;
    while ((ch = fgetc(fp)) != EOF) {
        putchar(ch);
    }
    printf("\n");
    fclose(fp);
    return 0;}
```

Task

- **Write a C program that performs the following tasks:**
  - Writes a given string to a file (get the string from the user).
  - Reads the string back from the file
  - Searches for a given character in the string

Task 1

- Writes a given string to a file (get the string from the user).

```c
#include <stdio.h>
#include <string.h>

int main() {
    char str[1000];
    char searchChar;
    FILE *file;
    // Task 1..Get string input from user
    printf("Enter a string to store in file: ");
    fgets(str, sizeof(str), stdin);
    str[strcspn(str, "\n")] = '\0'; // Remove trailing newline

    // Store string in file
    file = fopen("MyFile.txt", "w");
    if (!file) {
        printf("Error: Unable to create file \n");
        return 1;
    }

    fprintf(file, "%s", str);
    fclose(file);
    printf("String successfully saved to \n");

```

Task 2

- Reads the string back from the file

```
38      //Task 2 .. Retrieve string from file
39      file = fopen("MyFile.txt", "r");
40      if (!file) {
41          printf("Error: Unable to read file\n");
42          return 1;
43      }
44
45      char fileContent[1000];
46      fgets(fileContent, sizeof(fileContent), file);
47      fclose(file);
48      printf("Retrieved string: \"%s\"\n\n", fileContent);
49
```

Task

- Searches for a given character in the string

```
 4  // Function to count character occurrences
 5  int countCharacterOccurrences(const char *str, char target) {
 6      int count = 0;
 7      for (int i = 0; str[i] != '\0'; i++) {
 8          if (str[i] == target) {
 9              count++;
10          }
11      }
12      return count;
13  }
```

```
56      //Task 3 .. Get search character from user
57      printf("Enter character to count: ");
58      scanf(" %c", &searchChar);
59
60      // Count and display occurrences
61      int occurrences = countCharacterOccurrences(fileContent, searchChar);
62      printf("\nThe character '%c' appears %d time%s in the string.\n",
63              searchChar, occurrences, (occurrences != 1) ? "s" : "");
64
65      return 0;
66  }
```